

LiionTR

A Modular Scientific Framework for Lithium-Ion Battery Thermal Runaway Modeling

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Abstract

LiionTR is a modular scientific Python framework for the simulation of lithium-ion battery thermal runaway. The framework separates thermal runaway reaction kinetics, heat generation, gas generation, internal pressure, venting, and numerical integration into independent physical components.

The present formulation uses a lumped thermal model coupled to a network of reduced-order thermal runaway reactions. Gas generation, ideal-gas pressurization, and compressible vent flow can be incorporated into the same numerical problem. The architecture is designed to support future integration with thermochemical equilibrium calculations through Cantera and electrochemical models through PyBaMM.

This report describes the mathematical formulation, software architecture, implemented thermal runaway models, numerical methods, verification strategy, current limitations, and planned extensions of LiionTR.

Contents

Nomenclature	6
1 Introduction	8
1.1 Project objectives	8
1.2 Design philosophy	9
1.3 Reduced-order modeling	9
1.4 Current implementation	10
1.5 Relationship to external scientific software	10
1.6 Scope of this report	11
2 Physical and Mathematical Framework	11
2.1 Thermal state	12
2.2 Reaction states	12
2.3 Heat generation	12
2.4 Gas-generation states	12

2.5	Pressure state	13
2.6	Model coupling	13
2.7	Separation of physical models and state integration	13
2.8	Scope of the present framework	15
3	Thermal Model	15
3.1	Cell thermal capacity	15
3.2	Convective heat transfer	16
3.3	Internal heat generation	16
3.4	Thermal runaway feedback	16
3.5	Validity of the lumped approximation	16
3.6	Current assumptions	17
3.7	Limitations and future extensions	17
4	Reaction Kinetics	18
4.1	Reaction conversion	18
4.2	Arrhenius kinetics	18
4.3	Power-law progress	18
4.4	Autocatalytic progress	19
4.5	Temperature thresholds	19
4.6	Context-dependent reaction progress	19
4.7	Threshold progress	19
4.8	Exponential inhibition	20
4.9	Normalized context variables	20
4.10	Separation of kinetics and progress	20
4.11	Current limitations	21
5	Reaction Network	21
5.1	Standard reaction model	21
5.2	Reaction enthalpy convention	22
5.3	Reaction mass fraction	22
5.4	Reaction context	22
5.5	Multi-channel reactions	22
5.6	Shared reaction progress	23
5.7	Reaction network	23
5.8	Network heat generation	23
5.9	Coupled thermal-reaction system	24
5.10	Initial reaction state	24
5.11	Software representation	24
5.12	Current assumptions	24
5.13	Future extensions	25

6	Hu et al. (2020) Thermal Runaway Model	26
6.1	Reaction-state vector	26
6.2	Arrhenius formulation	26
6.3	SEI decomposition	27
6.4	Anode-electrolyte reaction	27
6.4.1	Coupling to SEI decomposition	28
6.5	Cathode decomposition	28
6.5.1	Cathode channel 1	29
6.5.2	Cathode channel 2	29
6.5.3	Shared cathode progress	29
6.6	Electrolyte decomposition	29
6.7	Summary of model parameters	30
6.8	Volumetric content and effective mass fractions	30
6.9	Complete thermal coupling	31
6.10	LiionTR representation	31
6.11	Scope of the implementation	32
7	Gas Generation	32
7.1	Gas species	32
7.2	Gas inventory	33
7.3	Mean molar mass	33
7.4	Empirical reaction gas yields	33
7.5	Total species generation rate	34
7.6	Cumulative gas generation	34
7.7	Closed-cell gas balance	34
7.8	Gas balance after vent opening	35
7.9	Coupling to pressure	35
7.10	Coupling to venting	35
7.11	Current assumptions	35
7.12	Limitations of fixed species yields	36
7.13	Future thermochemical formulation	36
8	Pressure and Venting	37
8.1	Ideal-gas pressure	37
8.2	Initial gas inventory	37
8.3	Pressure-temperature coupling	37
8.4	Vent-opening criterion	38
8.5	Maximum-pressure criterion	38
8.6	Compressible vent flow	38
8.7	Specific gas constant	39
8.8	Heat-capacity ratio	39
8.9	Critical pressure ratio	39
8.10	Choked mass flow	39

8.11	Unchoked mass flow	39
8.12	Reverse flow	40
8.13	Molar vent flow	40
8.14	Species vent rates	40
8.15	Closed and open governing equations	40
8.16	Irreversible vent transition	40
8.17	Coupled pressure-vent feedback	41
8.18	Current assumptions	41
8.19	Limitations	42
8.20	Future extensions	42
9	Numerical Method	43
9.1	State vector	43
9.2	Initial conditions	43
9.3	Right-hand-side evaluation	43
9.4	Gas-generation contribution	44
9.5	Pressure reconstruction	44
9.6	Vent-opening event	44
9.7	Two-phase integration	45
9.8	Solver flow	45
9.9	Maximum-temperature event	45
9.10	Maximum-pressure event	45
9.11	Stiff numerical integration	46
9.12	Adaptive time stepping	46
9.13	State reconstruction	47
9.14	Numerical architecture	47
9.15	Current limitations	48
9.16	Future numerical extensions	48
10	Verification	48
10.1	Unit-level verification	49
10.2	Dimensional consistency	49
10.3	Reaction-energy consistency	50
10.4	Hu model energy check	50
10.5	Thermal-response consistency	50
10.6	Gas-generation conservation	51
10.7	Pressure verification	51
10.8	Vent-flow verification	51
10.9	Vented coupled simulation	51
10.10	Regression testing	52
10.11	Verification hierarchy	52
10.12	Experimental validation	52

11 Reference Simulation Results	53
11.1 Reference simulation conditions	54
11.2 Temperature evolution	54
11.3 Reaction progression	54
11.4 Reaction heat generation	54
11.5 Integrated reaction energy	54
12 Future Development	56
12.1 Thermochemical gas modeling	56
12.2 Cantera integration	56
12.3 Energy-consistent thermochemistry	57
12.4 Electrochemical coupling	57
12.5 Calorimetric parameter estimation	58
12.6 Sensitivity and uncertainty analysis	58
12.7 Temperature-dependent material properties	59
12.8 Spatial thermal models	59
12.9 Cell-to-cell propagation	59
12.10 Mechanical venting and rupture	59
12.11 Multiphase venting	60
12.12 External vent-jet and combustion models	60
12.13 Validation database	61
12.14 Software development	61
12.15 Long-term architecture	62

Nomenclature

Symbol	Description	Unit
A	Arrhenius pre-exponential factor	1/s
A_v	Effective vent flow area	m ²
A_{cell}	External cell surface area	m ²
C_d	Vent discharge coefficient	–
C_{th}	Effective cell thermal capacity	J/K
c_p	Specific heat capacity	J/(kg K)
c_v	Constant-volume specific heat capacity	J/(kg K)
E_a	Activation energy	J/mol
f	Reaction-progress function	–
h	Convective heat-transfer coefficient	W/(m ² K)
$k(T)$	Temperature-dependent kinetic rate coefficient	1/s
k	Effective thermal conductivity	W/(m K)
L_c	Characteristic thermal length	m
m_{cell}	Battery-cell mass	kg
m_{gas}	Total gas mass	kg
$\dot{m}_{\text{vent}}$	Vent mass-flow rate	kg/s
M_s	Molar mass of gas species s	kg/mol
$\overline{M}$	Mean molar mass of the gas mixture	kg/mol
n_s	Mole inventory of gas species s	mol
n_{tot}	Total gas mole inventory	mol
$\dot{n}_{s,\text{gen}}$	Generation rate of gas species s	mol/s
$\dot{n}_{s,\text{vent}}$	Vent rate of gas species s	mol/s
P	Absolute internal gas pressure	Pa
P_0	Initial pressure	Pa
P_d	Downstream pressure	Pa
P_u	Upstream pressure	Pa
P_{vent}	Vent-opening pressure	Pa
P_{max}	Maximum permitted pressure	Pa
$\dot{Q}_{\text{gen}}$	Total internal heat-generation rate	W
$\dot{Q}_{\text{loss}}$	Heat-loss rate	W
$\dot{q}$	Mass-specific heat-generation rate	W/kg
R	Universal gas constant	J/(mol K)
R_s	Specific gas constant	J/(kg K)
r	Remaining reaction fraction, $1 - \alpha$	–

Symbol	Description	Unit
r_p	Downstream-to-upstream pressure ratio	—
r_{crit}	Critical pressure ratio for choked flow	—
T	Lumped cell temperature	K
T_0	Initial cell temperature	K
T_∞	Ambient temperature	K
T_{th}	Reaction threshold temperature	K
T_{max}	Maximum permitted temperature	K
V	Cell volume	m ³
V_{free}	Internal free gas volume	m ³
W_i	Volumetric reacting-material content of reaction i	kg/m ³
w_i	Effective cell mass fraction associated with reaction i	—
x_s	Mole fraction of gas species s	—
$Y_{i,s}$	Yield of gas species s from reaction i	mol/kg
α_i	Conversion of thermal runaway reaction i	—
$\alpha_{i,0}$	Initial conversion of reaction i	—
γ	Gas heat-capacity ratio, c_p/c_v	—
ΔH_i	Effective specific heat release of reaction i	J/kg
ρ	Effective cell density	kg/m ³
$\boldsymbol{\alpha}$	Vector of reaction conversions	—
$\mathbf{n}$	Vector of gas-species mole inventories	mol
$\mathbf{y}$	Numerical state vector	—
Bi	Biot number	—

1 Introduction

Lithium-ion batteries are used extensively in transportation, portable electronics, aerospace systems, stationary energy storage, and numerous other applications requiring high energy density and rechargeable electrochemical storage.

Despite their advantages, lithium-ion cells can undergo thermal runaway when subjected to sufficiently severe thermal, electrical, mechanical, or internal abuse conditions. Thermal runaway is characterized by a self-accelerating sequence of exothermic reactions that can produce rapid temperature increase, gas generation, internal pressurization, venting, fire, and, under some conditions, propagation to neighbouring cells.

The physical processes involved span several strongly coupled domains, including

- electrochemical state evolution;
- decomposition reaction kinetics;
- heat generation and heat transfer;
- gas production;
- thermochemical gas-phase reactions;
- internal pressure evolution;
- venting and structural failure;
- external combustion;
- cell-to-cell thermal propagation.

A single mathematical model rarely represents all of these phenomena with the same level of fidelity. Thermal runaway simulations therefore often rely on reduced-order models calibrated against calorimetric, abuse-test, or literature data.

LiionTR is being developed as a modular scientific framework for constructing and solving such models while preserving a clear separation between individual physical mechanisms.

1.1 Project objectives

The primary objective of LiionTR is to provide a reusable computational framework for lithium-ion battery thermal runaway research.

The framework is designed to support:

- multiple thermal runaway reaction mechanisms;
- reusable kinetic and reaction-progress laws;
- configurable battery-cell geometries and materials;
- thermal energy balances;

- empirical and future thermochemical gas-generation models;
- internal gas-pressure calculations;
- compressible venting;
- numerical event handling;
- model verification and validation;
- future cell-to-cell propagation models.

1.2 Design philosophy

A central design principle is the separation of physical models from numerical integration.

For example, an Arrhenius model defines a temperature-dependent kinetic rate but does not know how the corresponding reaction state is integrated.

Similarly, a pressure model evaluates pressure from thermodynamic state variables but does not advance those variables in time.

The numerical solver acts as the orchestration layer that assembles these individual components into the complete governing system.

This separation is intended to improve

- physical transparency;
- software maintainability;
- testability;
- reuse of scientific models;
- comparison between alternative formulations.

1.3 Reduced-order modeling

The current LiionTR formulation is primarily zero-dimensional.

A battery cell is represented using a lumped temperature state and a set of scalar reaction-progress variables.

The thermal runaway process is described conceptually by

$$\text{temperature} \rightarrow \text{reaction rates} \rightarrow \text{heat generation} \rightarrow \text{temperature}, \quad (1)$$

forming the fundamental positive feedback loop responsible for thermal runaway.

Additional coupling can be introduced through

$$\text{reaction progress} \rightarrow \text{gas generation} \rightarrow \text{pressure} \rightarrow \text{venting}. \quad (2)$$

These reduced-order models are computationally inexpensive compared with fully spatial electrochemical, thermal, mechanical, or computational fluid dynamics models. They are therefore useful for parameter studies, model development, sensitivity analysis, and safety-oriented system simulations.

1.4 Current implementation

The present framework includes:

- generic cell, geometry, and material representations;
- Arrhenius and threshold kinetic models;
- multiple reaction-progress formulations;
- context-dependent reaction coupling;
- standard and multi-channel reactions;
- reaction networks;
- a lumped thermal model;
- empirical reaction-based gas generation;
- species-resolved gas inventories;
- an ideal-gas pressure model;
- choked and unchoked compressible vent flow;
- irreversible vent-opening events;
- stiff numerical integration.

The framework also includes an implementation of the reduced-order thermal runaway reaction model reported by Hu et al. [1].

1.5 Relationship to external scientific software

LiionTR is not intended to reproduce capabilities that are already provided by specialized scientific software.

Instead, external scientific packages may eventually be incorporated as backends.

Two particularly relevant examples are Cantera and PyBaMM.

Cantera provides thermodynamic, chemical-kinetic, and transport capabilities and is a candidate backend for equilibrium gas composition and temperature-dependent mixture properties [2].

PyBaMM provides a framework for physics-based battery modeling and may supply electrochemical states ranging from reduced-order models to porous-electrode formulations [3].

The intended long-term relationship is therefore

$$\text{PyBaMM} \rightarrow \text{electrochemical state}, \quad (3)$$

$$\text{Cantera} \rightarrow \text{thermochemical state}, \quad (4)$$

and

$$\text{LiionTR} \rightarrow \text{thermal runaway and safety coupling.} \quad (5)$$

1.6 Scope of this report

This report documents the mathematical formulation and current software architecture of LiionTR.

The following sections describe:

1. the general physical and mathematical framework;
2. the lumped thermal formulation;
3. reaction kinetics and progress models;
4. reaction-network construction;
5. implementation of the Hu et al. thermal runaway model;
6. gas generation;
7. pressure and venting;
8. numerical integration;
9. verification strategy;
10. planned future development.

The report describes the current state of the framework and therefore distinguishes between implemented capabilities and planned future extensions.

2 Physical and Mathematical Framework

LiionTR is organized around a modular formulation in which physical models are separated from numerical integration. The objective is to represent lithium-ion battery thermal runaway using a set of independent but coupled subsystems describing reaction kinetics, heat generation, gas generation, pressure evolution, and venting.

The state of a zero-dimensional LiionTR simulation can be written conceptually as

$$\mathbf{y} = [T, \alpha_1, \dots, \alpha_N, n_1, \dots, n_M]^T, \quad (6)$$

where T is the lumped cell temperature, α_i is the conversion of thermal runaway reaction i , and n_s is the amount of gas species s contained in the cell free volume.

The governing system may therefore be expressed as

$$\frac{d\mathbf{y}}{dt} = \mathbf{F}(t, \mathbf{y}), \quad (7)$$

where the right-hand side is assembled from the thermal, reaction, gas-generation, pressure, and vent-flow models.

2.1 Thermal state

The cell temperature evolves according to an energy balance,

$$C_{\text{th}} \frac{dT}{dt} = \dot{Q}_{\text{gen}} - \dot{Q}_{\text{loss}}, \quad (8)$$

where C_{th} is the effective cell thermal capacity, $\dot{Q}_{\text{gen}}$ is the total internal heat-generation rate, and $\dot{Q}_{\text{loss}}$ is the heat-transfer rate from the cell to its environment.

2.2 Reaction states

Each thermal runaway reaction is represented using a scalar conversion,

$$0 \leq \alpha_i \leq 1. \quad (9)$$

The corresponding reaction rate is represented by

$$\frac{d\alpha_i}{dt} = k_i(T) f_i(\alpha_i, \alpha), \quad (10)$$

where $k_i(T)$ describes the intrinsic temperature dependence and f_i describes the dependence on reaction progress and, when required, the state of other reactions.

The complete reaction-state vector is

$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_N]^T. \quad (11)$$

2.3 Heat generation

For a reaction occupying an effective cell mass fraction w_i and releasing a specific enthalpy ΔH_i , the mass-specific heat-generation rate is written as

$$\dot{q}_i = w_i \Delta H_i \frac{d\alpha_i}{dt}. \quad (12)$$

The total heat-generation rate for a cell of mass m_{cell} is then

$$\dot{Q}_{\text{gen}} = m_{\text{cell}} \sum_{i=1}^N \dot{q}_i. \quad (13)$$

2.4 Gas-generation states

When gas generation is enabled, each gas species is represented by a mole inventory n_s .

Before vent opening,

$$\frac{dn_s}{dt} = \dot{n}_{s,\text{gen}}, \quad (14)$$

where $\dot{n}_{s,\text{gen}}$ is the molar gas-generation rate.

After vent opening,

$$\frac{dn_s}{dt} = \dot{n}_{s,\text{gen}} - \dot{n}_{s,\text{vent}}. \quad (15)$$

2.5 Pressure state

The current reduced-order formulation evaluates internal gas pressure using the ideal-gas equation,

$$P = \frac{n_{\text{tot}}RT}{V_{\text{free}}}, \quad (16)$$

where

$$n_{\text{tot}} = \sum_{s=1}^M n_s. \quad (17)$$

The current formulation assumes that the gas temperature is equal to the lumped cell temperature.

2.6 Model coupling

The principal physical couplings represented in LiionTR are

1. temperature controls thermal runaway reaction rates;
2. thermal runaway reactions release heat;
3. heat generation modifies the cell temperature;
4. reaction progress can generate gas;
5. the gas inventory and temperature determine internal pressure;
6. internal pressure can trigger vent opening;
7. vent flow removes gas from the internal inventory.

The reaction-thermal coupling forms the fundamental positive feedback mechanism responsible for thermal runaway.

A conceptual representation of the current LiionTR architecture is shown in Figure 1.

2.7 Separation of physical models and state integration

A central design principle of LiionTR is that physical model objects do not advance their own numerical state.

Instead, the numerical solver owns the complete state vector and passes the current state to the physical models whenever the right-hand side is evaluated.

This design provides several advantages:

- deterministic evaluation of the governing equations;
- compatibility with standard ordinary differential equation solvers;
- reduced hidden coupling between physical components;
- independent verification of individual physical models;
- easier implementation of event handling and future alternative numerical solvers.

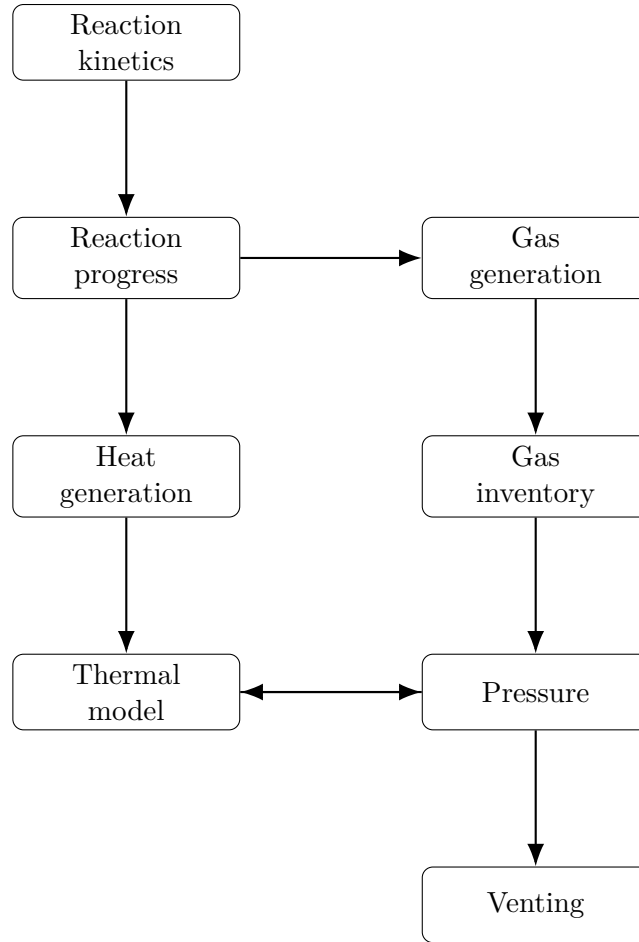


Figure 1: Conceptual architecture of the current LiionTR framework. Reaction kinetics determine reaction progress, which drives both heat generation and gas generation. Heat generation affects the thermal model, while gas generation affects the gas inventory and therefore the internal pressure. Pressure may eventually trigger venting.

2.8 Scope of the present framework

The current implementation focuses on reduced-order, zero-dimensional thermal runaway modeling.

The framework is intended to evolve toward more comprehensive physical descriptions, including thermochemical equilibrium, detailed electrochemical state information, spatially resolved heat transfer, cell-to-cell propagation, structural failure, and external vent-jet or combustion models.

Cantera is considered as a future thermochemical backend, while PyBaMM may provide electrochemical states and normal-operation battery physics. LiionTR is intended to remain the orchestration layer responsible for thermal runaway and battery-safety-specific coupling.

3 Thermal Model

The current LiionTR thermal formulation uses a zero-dimensional lumped-capacitance model. The complete battery cell is represented by a single temperature state, such that internal spatial temperature gradients are neglected.

The thermal energy balance is

$$C_{\text{th}} \frac{dT}{dt} = \dot{Q}_{\text{gen}} - hA(T - T_{\infty}), \quad (18)$$

where C_{th} is the cell thermal capacity, $\dot{Q}_{\text{gen}}$ is the total internal heat-generation rate, h is the convective heat-transfer coefficient, A is the external cell surface area, and T_{∞} is the ambient temperature.

The resulting temperature derivative is

$$\frac{dT}{dt} = \frac{\dot{Q}_{\text{gen}} - hA(T - T_{\infty})}{C_{\text{th}}}. \quad (19)$$

3.1 Cell thermal capacity

The effective thermal capacity is calculated as

$$C_{\text{th}} = m_{\text{cell}} c_p, \quad (20)$$

where c_p is the effective specific heat capacity of the cell.

The cell mass is calculated from the effective density and geometry,

$$m_{\text{cell}} = \rho V, \quad (21)$$

leading to

$$C_{\text{th}} = \rho V c_p. \quad (22)$$

In the present implementation, density and specific heat capacity are evaluated at 298.15 K when cell mass and thermal capacity are constructed. These quantities are therefore constant during a simulation.

3.2 Convective heat transfer

Heat exchange with the environment is currently represented using Newton's law of cooling,

$$\dot{Q}_{\text{conv}} = hA(T - T_{\infty}). \quad (23)$$

When $T > T_{\infty}$, this term removes heat from the cell.

If $T < T_{\infty}$, the same expression becomes negative and therefore represents heating of the cell by the surrounding environment.

3.3 Internal heat generation

Internal heat generation is supplied to the thermal model by the configured chemistry backend.

For a reaction network containing N thermal runaway reactions,

$$\dot{Q}_{\text{gen}} = m_{\text{cell}} \sum_{i=1}^N w_i \Delta H_i \dot{\alpha}_i. \quad (24)$$

The thermal model is therefore independent from the detailed kinetic mechanism. Different reaction networks can be coupled to the same thermal formulation.

3.4 Thermal runaway feedback

The coupling between reaction kinetics and temperature introduces a positive feedback mechanism.

For Arrhenius kinetics,

$$k(T) = A \exp\left(-\frac{E_a}{RT}\right). \quad (25)$$

As temperature rises, the reaction rate generally increases. This increases heat generation, which produces further temperature increase.

Conceptually,

$$T \uparrow \implies k(T) \uparrow \implies \dot{Q}_{\text{gen}} \uparrow \implies T \uparrow. \quad (26)$$

This feedback constitutes the principal mechanism of thermal runaway in the present reduced-order framework.

3.5 Validity of the lumped approximation

The lumped-capacitance approximation assumes that the internal temperature field equilibrates sufficiently rapidly compared with heat transfer across the external boundary.

A conventional criterion is the Biot number,

$$\text{Bi} = \frac{hL_c}{k}, \quad (27)$$

where L_c is a characteristic cell length and k is the effective thermal conductivity.

The lumped approximation is generally most appropriate when

$$\text{Bi} \ll 1. \tag{28}$$

LiionTR does not currently enforce a Biot-number criterion. Consequently, the validity of the lumped approximation must be assessed for the geometry, boundary conditions, and thermal runaway regime being simulated.

3.6 Current assumptions

The present thermal formulation assumes:

- a spatially uniform cell temperature;
- constant effective thermophysical properties;
- constant ambient temperature;
- uniform convection over the complete external cell surface;
- no radiative heat transfer;
- no explicit conductive coupling to neighbouring cells or structures;
- no explicit phase-change enthalpy;
- no internal thermal gradients.

3.7 Limitations and future extensions

A single-temperature model cannot reproduce internal hot spots or temperature gradients that may develop during severe thermal runaway.

Future thermal-model developments may therefore include:

- temperature-dependent density, heat capacity, and thermal conductivity;
- radiative heat transfer;
- conductive coupling between cells;
- one-dimensional or multidimensional thermal models;
- cell-to-cell thermal propagation;
- latent-heat and phase-change effects;
- energy coupling to vented material and gas enthalpy.

4 Reaction Kinetics

Thermal runaway reactions are represented using reduced-order kinetic models that relate reaction progress to temperature and reaction state.

For reaction i , the general progress equation is

$$\frac{d\alpha_i}{dt} = k_i(T) f_i(\alpha_i, \alpha), \quad (29)$$

where α_i is the dimensionless reaction conversion, $k_i(T)$ is the intrinsic temperature-dependent rate coefficient, and f_i is a progress function that may depend on the state of the complete reaction network.

4.1 Reaction conversion

Reaction progress is represented by a conversion variable satisfying

$$0 \leq \alpha \leq 1. \quad (30)$$

The unreacted or remaining fraction is therefore

$$r = 1 - \alpha. \quad (31)$$

An unreacted state corresponds to $\alpha = 0$, while a fully reacted state corresponds to $\alpha = 1$.

4.2 Arrhenius kinetics

The principal kinetic model currently implemented in LiionTR is the Arrhenius relation,

$$k(T) = A \exp\left(-\frac{E_a}{RT}\right), \quad (32)$$

where A is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, and T is the absolute temperature.

For first-order kinetics, A has units of 1/s, while E_a is expressed in J/mol.

The exponential temperature dependence is central to thermal runaway. A relatively modest increase in temperature can produce a large increase in reaction rate, thereby increasing heat generation and further accelerating the thermal response.

4.3 Power-law progress

A common progress model is based on the remaining reactant fraction,

$$f(\alpha) = (1 - \alpha)^n, \quad (33)$$

where n is the reaction order.

The complete progress equation becomes

$$\frac{d\alpha}{dt} = A \exp\left(-\frac{E_a}{RT}\right) (1 - \alpha)^n. \quad (34)$$

For $n = 1$, the reaction rate decreases linearly with the remaining reactant fraction.

4.4 Autocatalytic progress

LiionTR also supports autocatalytic progress laws of the general form

$$f(\alpha) = \alpha^m (1 - \alpha)^n, \quad (35)$$

where m and n control the dependence on reacted and unreacted material, respectively.

Such a formulation can represent reactions that accelerate as reaction products or active sites accumulate before eventually slowing as the reactant is depleted.

A frequently used special case is

$$f(\alpha) = \alpha (1 - \alpha). \quad (36)$$

4.5 Temperature thresholds

Some reduced-order models assume that a reaction is inactive below a prescribed onset temperature.

LiionTR represents this using a thresholded kinetic rate,

$$k_{\text{th}}(T) = \begin{cases} 0, & T < T_{\text{th}}, \\ k(T), & T \geq T_{\text{th}}. \end{cases} \quad (37)$$

This formulation is useful when reproducing empirical literature models that prescribe an explicit reaction onset temperature.

4.6 Context-dependent reaction progress

Thermal runaway reactions are not always independent.

The progress of one reaction may depend on the state of another reaction, such as the depletion of a protective layer or the formation of reactive products.

LiionTR therefore allows a progress model to depend on a reaction context containing the complete conversion vector,

$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_N]^T. \quad (38)$$

The progress equation can therefore be written more generally as

$$\frac{d\alpha_i}{dt} = k_i(T) f_i(\alpha_i, \boldsymbol{\alpha}). \quad (39)$$

4.7 Threshold progress

A context-dependent reaction may be activated only when another state variable satisfies a prescribed condition.

For a context variable x , a generic threshold progress model may be written as

$$f_{\text{th}} = \begin{cases} 0, & x > x_{\text{th}}, \\ f(\alpha), & x \leq x_{\text{th}}. \end{cases} \quad (40)$$

This formulation is used to represent sequential or conditionally activated thermal runaway mechanisms.

4.8 Exponential inhibition

LiionTR also provides an exponential inhibition mechanism,

$$f_{\text{inh}} = f(\alpha) \exp(-x), \quad (41)$$

where x is a context-dependent inhibition variable.

As x increases, the effective reaction rate decreases exponentially.

Such a formulation can represent the influence of a protective layer or other inhibiting material on a subsequent reaction.

4.9 Normalized context variables

A context variable may be constructed from the conversion or remaining fraction of another reaction.

For example, the remaining fraction of reaction j is

$$r_j = 1 - \alpha_j. \quad (42)$$

A normalized remaining-fraction variable may then be defined as

$$x_j = \frac{r_j}{r_{\text{ref}}}, \quad (43)$$

where r_{ref} is a prescribed reference value.

This provides a flexible mechanism for reproducing coupling terms reported in reduced-order thermal runaway models.

4.10 Separation of kinetics and progress

LiionTR deliberately separates the intrinsic kinetic law from the reaction-progress model.

The kinetic model determines how temperature affects the intrinsic rate,

$$T \longrightarrow k(T), \quad (44)$$

while the progress model determines how reaction state modifies that rate,

$$\alpha \longrightarrow f(\alpha, \alpha). \quad (45)$$

The complete progress rate is obtained only after combining both terms.

This separation permits a common kinetic model, such as Arrhenius kinetics, to be reused with multiple progress laws.

4.11 Current limitations

The present kinetic framework is intended primarily for reduced-order thermal runaway modeling.

Current limitations include:

- no explicit pressure dependence in the standard kinetic laws;
- no transport-limited reaction formulation;
- no explicit dependence on electrode potential or lithium concentration;
- no uncertainty propagation for kinetic parameters;
- no automatic kinetic parameter estimation from calorimetric data;
- no automatic competition between alternative reaction mechanisms.

Future developments may include parameter calibration using ARC or DSC data, uncertainty quantification, and coupling to electrochemical state variables supplied by PyBaMM.

5 Reaction Network

The reaction subsystem combines kinetic laws, reaction-progress models, reaction enthalpies, and reacting-material inventories into physical thermal runaway reactions.

Each reaction contributes both a progress rate and a heat-generation rate to the coupled system.

5.1 Standard reaction model

For a standard reaction, the progress rate is

$$\dot{\alpha} = k(T)f(\alpha, \boldsymbol{\alpha}), \quad (46)$$

where $k(T)$ is the kinetic rate and f is the reaction-progress function.

If the reaction occupies an effective mass fraction w of the total cell mass and releases a specific enthalpy ΔH , the mass-specific heat-generation rate relative to total cell mass is

$$\dot{q} = w\Delta H\dot{\alpha}. \quad (47)$$

The total heat-generation rate for a cell of mass m_{cell} is therefore

$$\dot{Q} = m_{\text{cell}}w\Delta H\dot{\alpha}. \quad (48)$$

5.2 Reaction enthalpy convention

The parameter ΔH represents the effective heat release per unit mass of reacting material.

For the reduced-order exothermic reactions currently implemented, ΔH is treated as a positive heat-release quantity.

The resulting heat-generation term is therefore positive when

$$\dot{\alpha} > 0. \quad (49)$$

This convention avoids embedding thermodynamic sign conventions directly into the thermal balance and is consistent with the current LiionTR reaction implementation.

5.3 Reaction mass fraction

The parameter w represents the effective fraction of total cell mass participating in a particular reaction.

For literature models that report a volumetric reactant content W_r in kg/m^3 , LiionTR converts this quantity to a cell mass fraction using

$$w = \frac{W_r}{\rho_{\text{cell}}}, \quad (50)$$

where ρ_{cell} is the effective cell density.

This conversion enables literature models reported on a volumetric basis to be used consistently within the cell-mass-based thermal formulation.

5.4 Reaction context

Individual reactions may depend on the state of other reactions.

LiionTR therefore constructs a reaction context containing the complete reaction state

$$\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_N]^T. \quad (51)$$

The context provides access to quantities such as

$$\alpha_j \quad (52)$$

and

$$1 - \alpha_j, \quad (53)$$

as well as transformed or normalized context variables.

This permits reaction coupling without storing mutable integration state inside individual reaction objects.

5.5 Multi-channel reactions

Some literature models represent one physical decomposition process using several parallel kinetic channels.

LiionTR represents these processes with a common conversion variable and multiple kinetic channels.

The shared reaction progress rate is obtained from the sum of the individual kinetic channels,

$$\dot{\alpha} = f(\alpha, \boldsymbol{\alpha}) \sum_{j=1}^{N_c} k_j(T). \quad (54)$$

For N_c channels sharing a conversion α , the heat-generation rate can be written conceptually as

$$\dot{q} = w f(\alpha, \boldsymbol{\alpha}) \sum_{j=1}^{N_c} \Delta H_j k_j(T). \quad (55)$$

Each channel therefore contributes independently to heat generation, while the physical reaction retains only one integrated progress state.

5.6 Shared reaction progress

The use of a shared conversion is physically distinct from representing each kinetic channel as an independent reaction.

For a multi-channel decomposition mechanism, all channels consume the same effective reactant inventory.

The model therefore integrates

$$\alpha \quad (56)$$

rather than introducing separate conversions

$$\alpha_1, \alpha_2, \dots, \alpha_{N_c}. \quad (57)$$

This distinction is important for preserving a consistent reacting mass inventory.

5.7 Reaction network

A thermal runaway mechanism is assembled as a network containing multiple reactions.

For N reactions, the conversion vector is

$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_N]^T. \quad (58)$$

The corresponding progress-rate vector is

$$\dot{\boldsymbol{\alpha}} = [\dot{\alpha}_1, \dot{\alpha}_2, \dots, \dot{\alpha}_N]^T. \quad (59)$$

Each reaction is evaluated using the current temperature and current reaction-network state.

5.8 Network heat generation

The total mass-specific heat-generation rate is

$$\dot{q}_{\text{total}} = \sum_{i=1}^N \dot{q}_i. \quad (60)$$

The corresponding total cell heat-generation rate is

$$\dot{Q}_{\text{gen}} = m_{\text{cell}} \dot{q}_{\text{total}}. \quad (61)$$

This quantity is passed to the thermal model.

5.9 Coupled thermal-reaction system

The reaction network and thermal model form a strongly coupled nonlinear system,

$$\frac{dT}{dt} = \frac{m_{\text{cell}} \sum_{i=1}^N \dot{q}_i - hA(T - T_{\infty})}{C_{\text{th}}}, \quad (62)$$

with

$$\frac{d\alpha_i}{dt} = k_i(T) f_i(\alpha_i, \boldsymbol{\alpha}), \quad i = 1, \dots, N. \quad (63)$$

The reaction rates depend on temperature, while temperature depends on reaction heat generation.

This bidirectional coupling produces the positive feedback characteristic of thermal runaway.

5.10 Initial reaction state

Each reaction requires an initial conversion

$$\alpha_{i,0}. \quad (64)$$

The complete initial reaction state is

$$\boldsymbol{\alpha}_0 = [\alpha_{1,0}, \dots, \alpha_{N,0}]^T. \quad (65)$$

When a literature model reports an initial remaining fraction r_0 , LiionTR converts it using

$$\alpha_0 = 1 - r_0. \quad (66)$$

5.11 Software representation

The physical organization of the reaction subsystem can be summarized using the conceptual structure shown in Figure 2.

5.12 Current assumptions

The current reaction framework assumes that:

- every integrated reaction is represented by a scalar conversion;

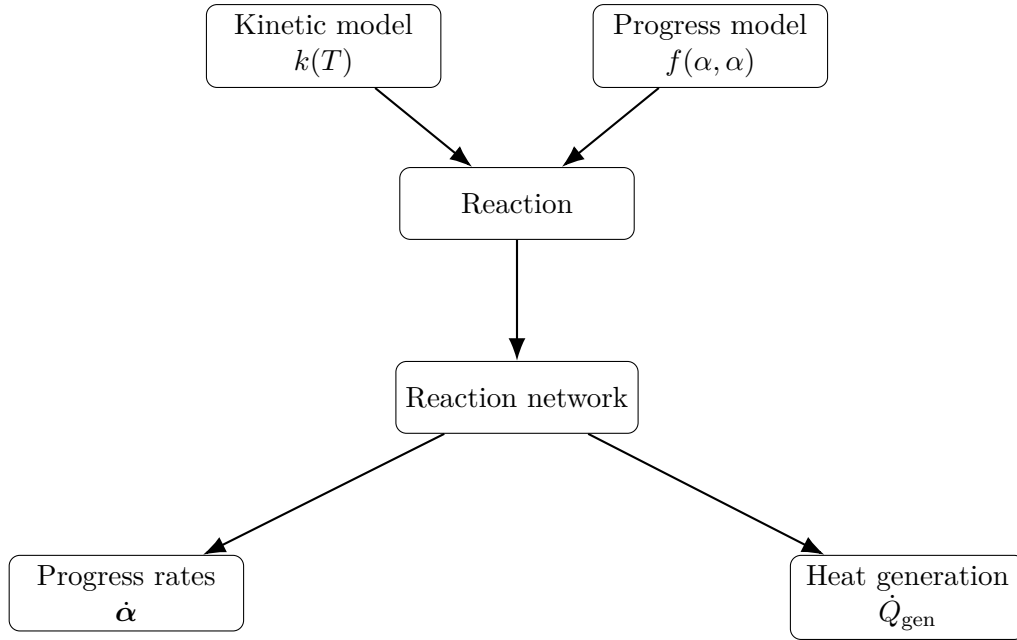


Figure 2: Conceptual organization of the LiionTR reaction subsystem. Kinetic and progress models are combined into reaction models, which are assembled into a reaction network that supplies reaction progress rates and total heat generation.

- reaction states are spatially uniform;
- reaction enthalpies are effective constant quantities;
- reacting-material mass fractions remain fixed;
- pressure dependence is absent unless explicitly introduced by a custom model;
- reaction products are not automatically determined from chemical equilibrium;
- transport limitations are not explicitly resolved.

5.13 Future extensions

The modular reaction architecture is intended to support more detailed physical descriptions.

Potential extensions include:

- temperature-dependent reaction enthalpies;
- pressure-dependent kinetics;
- transport-limited reactions;
- competing reaction mechanisms;
- reaction-derived elemental source inventories;
- thermochemical equilibrium using Cantera;
- electrochemical-state-dependent kinetics;

- calibration against ARC and DSC measurements;
- parameter uncertainty and sensitivity analysis.

6 Hu et al. (2020) Thermal Runaway Model

The first complete thermal runaway reaction network implemented in LiionTR is based on the reduced-order model reported by Hu et al. [1].

The model represents thermal runaway using four principal reaction states:

1. solid-electrolyte interphase (SEI) decomposition;
2. anode-electrolyte reaction;
3. cathode decomposition;
4. electrolyte decomposition.

Rather than implementing the model as a single monolithic component, LiionTR represents each physical contribution using the generic kinetic, progress, reaction, and reaction-network abstractions described in [sections 4](#) and [5](#).

6.1 Reaction-state vector

The reaction-state vector is

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_{\text{SEI}} \\ \alpha_{\text{an}} \\ \alpha_{\text{cat}} \\ \alpha_{\text{el}} \end{bmatrix}, \quad (67)$$

where the subscripts denote SEI decomposition, anode-electrolyte reaction, cathode decomposition, and electrolyte decomposition, respectively.

The initial state used by the current LiionTR implementation is

$$\boldsymbol{\alpha}_0 = \begin{bmatrix} 0.85 \\ 0.25 \\ 0.04 \\ 0.00 \end{bmatrix}. \quad (68)$$

6.2 Arrhenius formulation

The thermally activated reactions use the Arrhenius expression

$$k_i(T) = A_i \exp\left(-\frac{E_{a,i}}{RT}\right), \quad (69)$$

where A_i is the pre-exponential factor and $E_{a,i}$ is the activation energy of reaction i .

The resulting conversion rate is obtained by multiplying the kinetic term by the appropriate reaction-progress model,

$$\dot{\alpha}_i = k_i(T) f_i(\alpha_i, \boldsymbol{\alpha}). \quad (70)$$

6.3 SEI decomposition

The SEI decomposition reaction is represented using first-order remaining-fraction kinetics,

$$\frac{d\alpha_{\text{SEI}}}{dt} = k_{\text{SEI}}(T) (1 - \alpha_{\text{SEI}}). \quad (71)$$

The implemented kinetic parameters are

$$A_{\text{SEI}} = 1.667 \times 10^{15} \text{ 1/s}, \quad (72)$$

$$E_{a,\text{SEI}} = 1.3508 \times 10^5 \text{ J/mol}, \quad (73)$$

$$\Delta H_{\text{SEI}} = 2.57 \times 10^5 \text{ J/kg}. \quad (74)$$

The reported volumetric reactant content is converted to SI units as

$$6.104 \times 10^5 \text{ g/m}^3 = 610.4 \text{ kg/m}^3. \quad (75)$$

The literature model specifies an initial remaining fraction of 0.15. Because LiionTR uses conversion,

$$\alpha_{\text{SEI},0} = 1 - 0.15 = 0.85. \quad (76)$$

6.4 Anode-electrolyte reaction

The anode-electrolyte reaction uses the Arrhenius parameters

$$A_{\text{an}} = 2.5 \times 10^{13} \text{ 1/s}, \quad (77)$$

$$E_{a,\text{an}} = 1.3508 \times 10^5 \text{ J/mol}, \quad (78)$$

$$\Delta H_{\text{an}} = 1.714 \times 10^6 \text{ J/kg}. \quad (79)$$

The reaction enthalpy follows from the unit conversion

$$1714 \text{ J/g} = 1.714 \times 10^6 \text{ J/kg}. \quad (80)$$

The same volumetric reactant content as the SEI reaction is used,

$$W_{\text{an}} = 610.4 \text{ kg/m}^3. \quad (81)$$

The initial remaining fraction is 0.75, corresponding to

$$\alpha_{\text{an},0} = 1 - 0.75 = 0.25. \quad (82)$$

6.4.1 Coupling to SEI decomposition

The anode-electrolyte reaction is not represented as an independent first-order process. Its progress depends on the state of the SEI reaction.

The remaining SEI fraction is

$$r_{\text{SEI}} = 1 - \alpha_{\text{SEI}}. \quad (83)$$

In the current LiionTR implementation, the anode reaction becomes active only when

$$r_{\text{SEI}} < 0.10. \quad (84)$$

The reaction is additionally modified by an exponential inhibition term derived from the normalized SEI state.

The resulting model may be written conceptually as

$$\frac{d\alpha_{\text{an}}}{dt} = k_{\text{an}}(T) (1 - \alpha_{\text{an}}) I_{\text{SEI}}, \quad (85)$$

where

$$I_{\text{SEI}} = \exp(-x_{\text{SEI}}) \quad (86)$$

when the SEI threshold condition in [equation \(84\)](#) is satisfied; otherwise the reaction progress rate is zero.

This formulation illustrates the context-dependent reaction architecture implemented in LiionTR.

6.5 Cathode decomposition

Cathode decomposition differs from the preceding reactions because it is represented using two parallel kinetic channels sharing a single reaction conversion,

$$\alpha_{\text{cat}}. \quad (87)$$

The cathode reaction is activated only above

$$T_{\text{cat}} = 393.15 \text{ K}. \quad (88)$$

The reported cathode content is

$$1.221 \times 10^6 \text{ g/m}^3, \quad (89)$$

which LiionTR converts to

$$W_{\text{cat}} = 1221 \text{ kg/m}^3. \quad (90)$$

6.5.1 Cathode channel 1

The first kinetic channel uses

$$A_{\text{cat},1} = 1.75 \times 10^9 \text{ 1/s}, \quad (91)$$

$$E_{a,\text{cat},1} = 1.1495 \times 10^5 \text{ J/mol}, \quad (92)$$

$$\Delta H_{\text{cat},1} = 7.7 \times 10^4 \text{ J/kg}. \quad (93)$$

6.5.2 Cathode channel 2

The second kinetic channel uses

$$A_{\text{cat},2} = 1.077 \times 10^{12} \text{ 1/s}, \quad (94)$$

$$E_{a,\text{cat},2} = 1.5888 \times 10^5 \text{ J/mol}, \quad (95)$$

$$\Delta H_{\text{cat},2} = 8.4 \times 10^4 \text{ J/kg}. \quad (96)$$

6.5.3 Shared cathode progress

The cathode reaction uses the autocatalytic progress function

$$f_{\text{cat}} = \alpha_{\text{cat}} (1 - \alpha_{\text{cat}}). \quad (97)$$

The shared cathode conversion therefore evolves according to

$$\frac{d\alpha_{\text{cat}}}{dt} = f_{\text{cat}} [k_{\text{cat},1}(T) + k_{\text{cat},2}(T)]. \quad (98)$$

The initial conversion is

$$\alpha_{\text{cat},0} = 0.04. \quad (99)$$

Both kinetic channels use the same progress variable, and their heat contributions are combined.

The resulting mass-specific heat-generation rate is

$$\dot{q}_{\text{cat}} = w_{\text{cat}} f_{\text{cat}} [\Delta H_{\text{cat},1} k_{\text{cat},1}(T) + \Delta H_{\text{cat},2} k_{\text{cat},2}(T)], \quad (100)$$

for temperatures satisfying the threshold condition in [equation \(88\)](#).

6.6 Electrolyte decomposition

Electrolyte decomposition is represented by an Arrhenius reaction activated above

$$T_{\text{el}} = 473.15 \text{ K}. \quad (101)$$

The kinetic and energetic parameters are

$$A_{\text{el}} = 5.14 \times 10^{25} \text{ 1/s}, \quad (102)$$

$$E_{a,\text{el}} = 2.74 \times 10^5 \text{ J/mol}, \quad (103)$$

$$\Delta H_{\text{el}} = 1.55 \times 10^5 \text{ J/kg}. \quad (104)$$

The reported reaction enthalpy is converted according to

$$155 \text{ J/g} = 1.55 \times 10^5 \text{ J/kg}. \quad (105)$$

The reported volumetric electrolyte content is

$$4.069 \times 10^5 \text{ g/m}^3, \quad (106)$$

which becomes

$$W_{\text{el}} = 406.9 \text{ kg/m}^3. \quad (107)$$

The initial remaining fraction is unity, giving

$$\alpha_{\text{el},0} = 0. \quad (108)$$

6.7 Summary of model parameters

The principal parameters used by the current LiionTR implementation are summarized in [table 2](#).

Table 2: Principal kinetic and energetic parameters used in the LiionTR implementation of the thermal runaway model of Hu et al. [1].

Reaction	A (1/s)	E_a (J/mol)	ΔH (J/kg)	α_0
SEI	1.667×10^{15}	1.3508×10^5	2.57×10^5	0.85
Anode-electrolyte	2.5×10^{13}	1.3508×10^5	1.714×10^6	0.25
Cathode channel 1	1.75×10^9	1.1495×10^5	7.7×10^4	0.04
Cathode channel 2	1.077×10^{12}	1.5888×10^5	8.4×10^4	0.04
Electrolyte	5.14×10^{25}	2.74×10^5	1.55×10^5	0.00

6.8 Volumetric content and effective mass fractions

Hu et al. report several reacting-material inventories on a volumetric basis [1].

LiionTR converts these quantities to effective mass fractions using

$$w_i = \frac{W_i}{\rho_{\text{cell}}}, \quad (109)$$

where W_i is the volumetric reactant content and ρ_{cell} is the effective cell density.

The total heat generation from reaction i is consequently

$$\dot{Q}_i = m_{\text{cell}} w_i \Delta H_i \dot{\alpha}_i. \quad (110)$$

6.9 Complete thermal coupling

The total heat-generation rate is

$$\dot{Q}_{\text{gen}} = \dot{Q}_{\text{SEI}} + \dot{Q}_{\text{an}} + \dot{Q}_{\text{cat}} + \dot{Q}_{\text{el}}. \quad (111)$$

This source term is coupled directly to the lumped thermal balance,

$$C_{\text{th}} \frac{dT}{dt} = \dot{Q}_{\text{gen}} - hA(T - T_{\infty}). \quad (112)$$

The Arrhenius temperature dependence then creates the thermal feedback responsible for runaway behavior.

6.10 LiionTR representation

The mapping of the Hu et al. model onto the LiionTR architecture is illustrated in [figure 3](#).

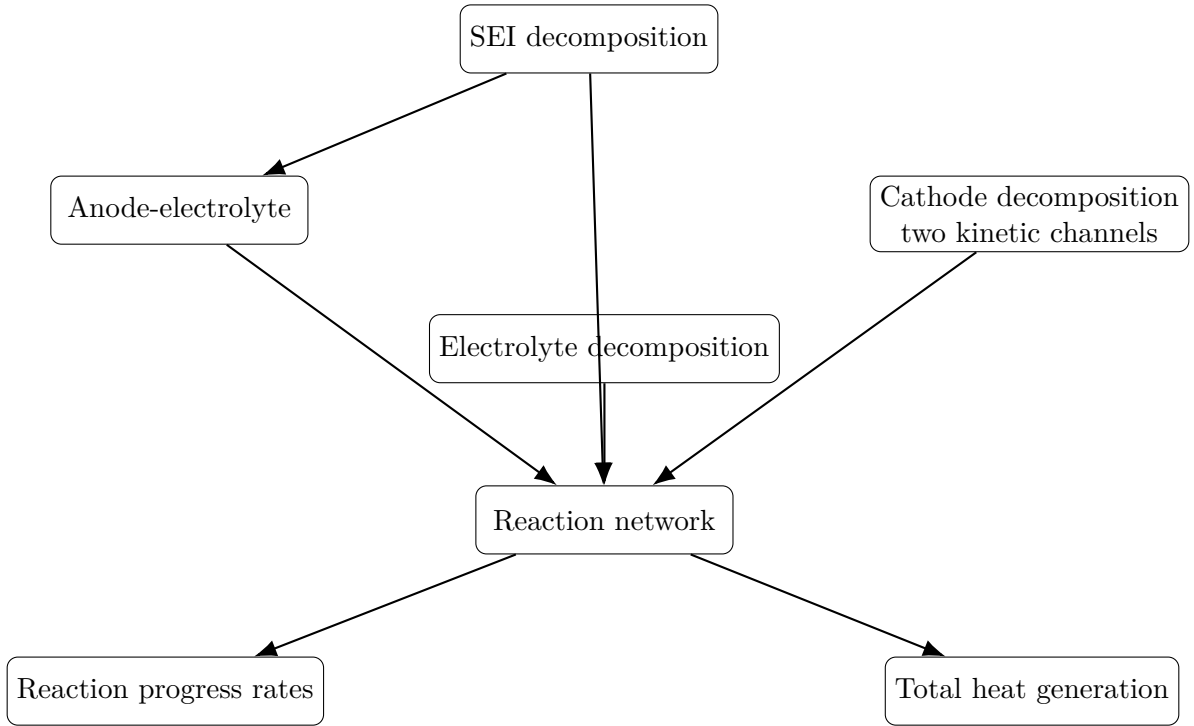


Figure 3: Representation of the Hu et al. thermal runaway model within the generic LiionTR reaction-network architecture.

The arrow between the SEI and anode blocks indicates that the anode-electrolyte reaction depends on the state of the SEI reaction. The remaining reactions are assembled into the same network and coupled through the common thermal state.

6.11 Scope of the implementation

The present implementation focuses on the reduced-order reaction and heat-generation structure described by Hu et al. [1].

The LiionTR representation should therefore be distinguished from a complete mechanistic electrochemical or thermochemical model.

In particular, the current Hu reaction network does not itself determine

- equilibrium gas composition;
- gas-phase secondary reactions;
- electrochemical electrode potentials;
- spatial temperature distributions;
- mechanical deformation or rupture;
- multiphase vent products.

These phenomena are intended to be represented by separate LiionTR subsystems or external scientific backends.

7 Gas Generation

Thermal runaway reactions can produce substantial quantities of gas. LiionTR currently represents gas generation using a reduced-order empirical formulation in which prescribed species yields are associated with reaction progress.

This formulation provides the coupling required to study gas inventory, internal pressure, and venting while keeping the gas-generation model independent from the reaction kinetics and numerical solver.

7.1 Gas species

Each gas species s is characterized by a molar mass

$$M_s, \tag{113}$$

expressed in kg/mol.

The instantaneous gas state is represented by the mole inventory

$$n_s, \tag{114}$$

for each species.

7.2 Gas inventory

For a mixture containing N_s species, the total gas amount is

$$n_{\text{tot}} = \sum_{s=1}^{N_s} n_s. \quad (115)$$

The total gas mass is

$$m_{\text{gas}} = \sum_{s=1}^{N_s} n_s M_s. \quad (116)$$

The mole fraction of species s is

$$x_s = \frac{n_s}{n_{\text{tot}}}, \quad (117)$$

such that

$$\sum_{s=1}^{N_s} x_s = 1 \quad (118)$$

for a nonempty gas inventory.

7.3 Mean molar mass

The current venting model uses the mole-weighted mean molar mass

$$\overline{M} = \sum_{s=1}^{N_s} x_s M_s. \quad (119)$$

Equivalently,

$$\overline{M} = \frac{\sum_{s=1}^{N_s} n_s M_s}{\sum_{s=1}^{N_s} n_s}. \quad (120)$$

This effective property provides the connection between the species-resolved gas inventory and the single-mixture compressible-flow model.

7.4 Empirical reaction gas yields

The current LiionTR gas-generation model assigns prescribed gas yields to individual thermal runaway reactions.

For reaction i and species s , the gas yield is denoted

$$Y_{i,s}, \quad (121)$$

with units of

$$\text{mol/kg} \quad (122)$$

of reacted material.

If reaction i occupies an effective mass fraction w_i of a cell with mass m_{cell} , the corresponding reacted-material rate is

$$\frac{dm_{i,\text{reacted}}}{dt} = m_{\text{cell}} w_i \frac{d\alpha_i}{dt}. \quad (123)$$

The generation rate of species s from reaction i is therefore

$$\dot{n}_{i,s} = Y_{i,s} m_{\text{cell}} w_i \dot{\alpha}_i. \quad (124)$$

7.5 Total species generation rate

When multiple reactions generate the same gas species, their contributions are summed.

The total generation rate of species s is

$$\dot{n}_{s,\text{gen}} = \sum_{i=1}^{N_r} Y_{i,s} m_{\text{cell}} w_i \dot{\alpha}_i, \quad (125)$$

where N_r is the number of thermal runaway reactions.

7.6 Cumulative gas generation

If reaction i evolves from an initial conversion $\alpha_{i,0}$ to α_i , the corresponding conversion change is

$$\Delta\alpha_i = \alpha_i - \alpha_{i,0}. \quad (126)$$

The cumulative gas amount associated with species s is then

$$n_{s,\text{generated}} = \sum_{i=1}^{N_r} Y_{i,s} m_{\text{cell}} w_i (\alpha_i - \alpha_{i,0}). \quad (127)$$

This expression is useful for closed-system calculations and for verification of the integrated gas-generation rates.

7.7 Closed-cell gas balance

Before vent opening, all generated gas remains inside the modeled free volume.

The species balance is

$$\frac{dn_s}{dt} = \dot{n}_{s,\text{gen}}. \quad (128)$$

The total mole inventory therefore increases according to

$$\frac{dn_{\text{tot}}}{dt} = \sum_s \dot{n}_{s,\text{gen}}. \quad (129)$$

7.8 Gas balance after vent opening

After vent opening, gas generation and gas discharge occur simultaneously.

The species balance becomes

$$\frac{dn_s}{dt} = \dot{n}_{s,\text{gen}} - \dot{n}_{s,\text{vent}}. \quad (130)$$

The instantaneous gas composition may therefore evolve because of both reaction-driven generation and removal through the vent.

7.9 Coupling to pressure

The gas inventory supplies the total gas amount used by the pressure model.

For the current ideal-gas formulation,

$$P = \frac{n_{\text{tot}}RT}{V_{\text{free}}}. \quad (131)$$

Thus, both gas generation and temperature evolution contribute directly to cell pressurization.

7.10 Coupling to venting

The gas inventory also provides the mixture composition required by the vent-flow model.

The mean molar mass is calculated using [equation \(119\)](#), and the total molar discharge rate is distributed between species according to their instantaneous mole fractions.

This coupling is represented schematically in [figure 4](#).

7.11 Current assumptions

The empirical gas-generation formulation assumes that:

- gas yields are prescribed directly for individual reactions;
- species yields remain constant during each reaction;
- gas yields do not explicitly depend on temperature or pressure;
- gas-phase secondary chemistry is not represented;
- the gas mixture is spatially homogeneous;
- all gas species share a common temperature;
- the gas temperature is equal to the lumped cell temperature;
- condensation and phase separation are neglected;
- gas dissolution into condensed phases is neglected.

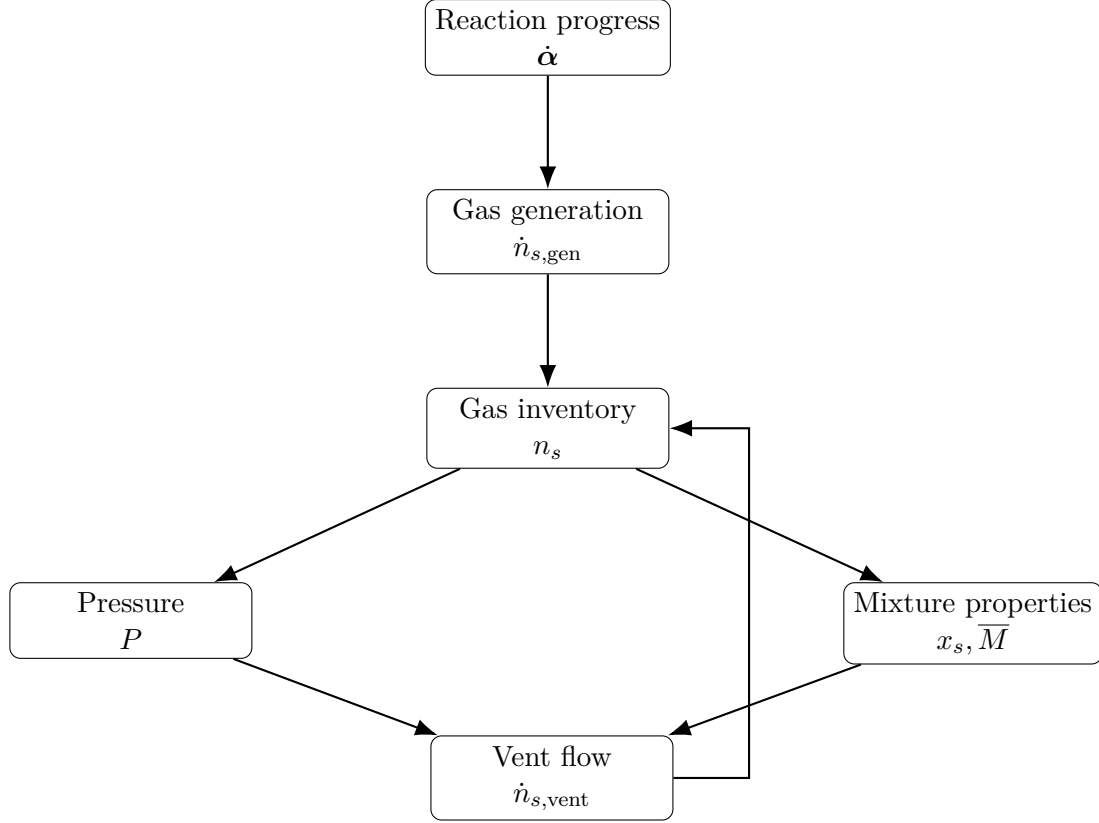


Figure 4: Coupling between reaction-driven gas generation, the gas inventory, pressure, mixture properties, and vent discharge in the current LiionTR formulation.

7.12 Limitations of fixed species yields

Thermal runaway gas composition can depend strongly on battery chemistry, electrolyte composition, state of charge, temperature, pressure, and secondary reactions.

Actual vent gases may contain, among other species,

$$\text{H}_2, \text{H}_2\text{O}, \text{CO}, \text{CO}_2, \text{CH}_4, \text{C}_2\text{H}_4. \quad (132)$$

Additional fluorinated or phosphorus-containing species may also be important depending on electrolyte chemistry.

A fixed species-yield model cannot generally predict these composition changes from first principles.

7.13 Future thermochemical formulation

The intended long-term thermochemical architecture is to replace or complement direct species yields with elemental source inventories.

Conceptually,

$$\text{reaction progress} \longrightarrow \text{element inventory} \longrightarrow \text{chemical equilibrium} \longrightarrow \text{gas composition}. \quad (133)$$

Relevant elemental inventories may include

$$\text{C, H, O, N, F, P, Li.} \quad (134)$$

A thermochemical backend such as Cantera could then provide equilibrium composition and mixture properties.

The present empirical implementation is retained because it provides a simple, deterministic, and independently testable formulation while the thermochemical architecture is developed.

8 Pressure and Venting

The gas-generation subsystem is coupled to internal pressure and venting models in order to represent cell pressurization and subsequent gas discharge.

The current formulation combines an ideal-gas pressure model with compressible single-phase vent flow.

8.1 Ideal-gas pressure

The cell gas pressure is calculated using

$$PV_{\text{free}} = n_{\text{tot}}RT, \quad (135)$$

or

$$P = \frac{n_{\text{tot}}RT}{V_{\text{free}}}, \quad (136)$$

where V_{free} is the available gas volume.

All pressures are treated as absolute pressures.

8.2 Initial gas inventory

The initial gas amount is calculated from the specified initial pressure P_0 , initial temperature T_0 , and free volume:

$$n_0 = \frac{P_0 V_{\text{free}}}{RT_0}. \quad (137)$$

Before venting, the total amount of gas can be represented as the sum of the initial inventory and generated gas.

During venting, however, pressure must be reconstructed directly from the instantaneous integrated species inventories.

8.3 Pressure-temperature coupling

For fixed volume,

$$P \propto n_{\text{tot}}T. \quad (138)$$

Cell pressure can therefore rise because of both

1. gas generation, which increases n_{tot} ;
2. thermal runaway, which increases T .

The pressure response is thus strongly coupled to the thermal and reaction subsystems.

8.4 Vent-opening criterion

A vent-opening pressure

$$P_{\text{vent}} \tag{139}$$

may be specified as part of the thermal problem.

The vent-opening event occurs when

$$P = P_{\text{vent}}. \tag{140}$$

Before this condition is reached, the gas system is treated as closed.

After the event, the vent is considered permanently open.

8.5 Maximum-pressure criterion

An independent maximum-pressure threshold may also be specified,

$$P_{\text{max}}. \tag{141}$$

This condition may be used to terminate numerical integration when

$$P = P_{\text{max}}. \tag{142}$$

The maximum-pressure threshold is distinct from the vent-opening pressure and may represent a numerical safety criterion or a simplified containment-failure limit.

8.6 Compressible vent flow

Gas discharge through the vent is represented using compressible ideal-gas flow.

The upstream pressure is

$$P_u = P_{\text{cell}}, \tag{143}$$

while the downstream pressure is typically the ambient pressure,

$$P_d = P_{\text{ambient}}. \tag{144}$$

The pressure ratio is

$$r_p = \frac{P_d}{P_u}. \tag{145}$$

8.7 Specific gas constant

The specific gas constant of the mixture is calculated as

$$R_s = \frac{R}{\overline{M}}, \quad (146)$$

where $\overline{M}$ is the instantaneous mean molar mass of the gas mixture.

8.8 Heat-capacity ratio

The current vent model uses a prescribed heat-capacity ratio

$$\gamma = \frac{c_p}{c_v}. \quad (147)$$

In the present implementation, γ is treated as constant rather than being recomputed from the instantaneous gas composition.

8.9 Critical pressure ratio

For ideal-gas isentropic flow, the critical downstream-to-upstream pressure ratio is

$$r_{\text{crit}} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}}. \quad (148)$$

The flow is considered choked when

$$\frac{P_d}{P_u} \leq r_{\text{crit}}. \quad (149)$$

8.10 Choked mass flow

For choked flow, the mass discharge rate is

$$\dot{m}_{\text{vent}} = C_d A_v P_u \sqrt{\frac{\gamma}{R_s T}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma + 1}{2(\gamma - 1)}}. \quad (150)$$

where C_d is the discharge coefficient and A_v is the effective vent area.

Once the flow is choked, decreasing downstream pressure further does not increase the mass flow predicted by this relation.

8.11 Unchoked mass flow

When

$$\frac{P_d}{P_u} > r_{\text{crit}}, \quad (151)$$

the discharge is treated as unchoked.

The mass flow rate is

$$\dot{m}_{\text{vent}} = C_d A_v P_u \sqrt{\frac{2\gamma}{R_s T (\gamma - 1)} \left[\left(\frac{P_d}{P_u} \right)^{2/\gamma} - \left(\frac{P_d}{P_u} \right)^{(\gamma+1)/\gamma} \right]}. \quad (152)$$

As P_d approaches P_u , the predicted discharge rate approaches zero.

8.12 Reverse flow

The current formulation does not represent inward flow through the vent.

Consequently,

$$P_u \leq P_d \implies \dot{m}_{\text{vent}} = 0. \quad (153)$$

8.13 Molar vent flow

The total mass discharge rate is converted to a molar flow rate using

$$\dot{n}_{\text{vent}} = \frac{\dot{m}_{\text{vent}}}{M}. \quad (154)$$

8.14 Species vent rates

The current model assumes a well-mixed gas inventory.

The vented gas therefore has the same mole fractions as the gas remaining inside the modeled free volume.

For species s ,

$$\dot{n}_{s,\text{vent}} = x_s \dot{n}_{\text{vent}}. \quad (155)$$

Summing over species gives

$$\sum_s \dot{n}_{s,\text{vent}} = \dot{n}_{\text{vent}}. \quad (156)$$

8.15 Closed and open governing equations

The vent model creates two distinct gas-balance phases.

Before vent opening,

$$\frac{dn_s}{dt} = \dot{n}_{s,\text{gen}}. \quad (157)$$

After vent opening,

$$\frac{dn_s}{dt} = \dot{n}_{s,\text{gen}} - \dot{n}_{s,\text{vent}}. \quad (158)$$

8.16 Irreversible vent transition

The current LiionTR implementation treats vent opening as an irreversible state transition,

$$\text{CLOSED} \longrightarrow \text{OPEN}. \quad (159)$$

The vent does not reseal if pressure subsequently falls below P_{vent} .

The numerical interpretation of this transition is illustrated in [figure 5](#).

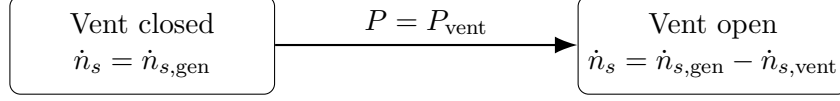


Figure 5: Irreversible vent-state transition used by the current LiionTR solver.

8.17 Coupled pressure-vent feedback

After vent opening, pressure determines the vent-flow rate, while the vent-flow rate modifies the gas inventory and therefore pressure.

This feedback may be represented as

$$P \longrightarrow \dot{n}_{\text{vent}} \longrightarrow n_{\text{tot}} \longrightarrow P. \quad (160)$$

Gas generation and temperature continue to evolve simultaneously.

The pressure response therefore depends on the competition between

$$\dot{n}_{\text{gen}} \quad (161)$$

and

$$\dot{n}_{\text{vent}}, \quad (162)$$

together with the evolving thermal state.

8.18 Current assumptions

The pressure and venting formulation assumes:

- ideal-gas behavior;
- constant free volume;
- uniform gas pressure;
- gas temperature equal to the lumped cell temperature;
- single-phase gas discharge;
- quasi-steady compressible vent flow;
- fixed effective vent area;
- constant discharge coefficient;

- constant heat-capacity ratio;
- a well-mixed gas inventory;
- no preferential species transport through the vent;
- no reverse flow;
- irreversible vent opening.

8.19 Limitations

Real lithium-ion battery venting can involve considerably more complex behavior than represented by the present model.

The discharged material may contain

- permanent gases;
- electrolyte vapor;
- liquid droplets;
- aerosols;
- condensed particles;
- decomposed electrode and separator material.

The physical vent may also deform progressively, resulting in a time-dependent effective flow area.

These mechanisms are outside the scope of the present reduced-order model.

8.20 Future extensions

Future developments may include:

- gas properties obtained dynamically from Cantera;
- temperature- and composition-dependent γ ;
- variable vent geometry;
- progressive vent opening;
- mechanical rupture models;
- two-phase and multiphase discharge;
- electrolyte flashing;
- vent-jet momentum models;
- combustion and fire models downstream of the vent.

9 Numerical Method

LiionTR formulates thermal runaway as a coupled system of ordinary differential equations integrated in time.

The numerical solver owns the complete dynamic state and evaluates the physical models only as functions of the current state. This avoids hidden mutable state inside the thermal, reaction, gas, pressure, and venting models.

9.1 State vector

For a problem containing N_r reaction states and N_s gas species, the integrated state vector is

$$\mathbf{y} = \begin{bmatrix} T & \alpha_1 & \cdots & \alpha_{N_r} & n_1 & \cdots & n_{N_s} \end{bmatrix}^T. \quad (163)$$

The first component is the lumped cell temperature.

The following N_r components contain reaction conversions,

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 & \cdots & \alpha_{N_r} \end{bmatrix}^T, \quad (164)$$

while the final N_s components contain the gas-species inventories,

$$\mathbf{n} = \begin{bmatrix} n_1 & \cdots & n_{N_s} \end{bmatrix}^T. \quad (165)$$

When gas generation is disabled, the gas portion of the state vector is absent.

9.2 Initial conditions

The initial numerical state is assembled from the configured thermal problem.

The temperature is initialized as

$$T(0) = T_0. \quad (166)$$

Each reaction is initialized according to its prescribed initial conversion,

$$\alpha_i(0) = \alpha_{i,0}. \quad (167)$$

When gas modeling is enabled, the initial gas inventory is determined from the configured species inventory or from the pressure model.

The complete initial state is therefore

$$\mathbf{y}(0) = \mathbf{y}_0. \quad (168)$$

9.3 Right-hand-side evaluation

At every solver evaluation, the current state vector is unpacked into temperature, reaction conversions, and gas inventories.

The reaction context is constructed from the current conversion vector,

$$\boldsymbol{\alpha}(t). \quad (169)$$

The reaction network then evaluates

$$\dot{\boldsymbol{\alpha}} = \mathbf{f}_{\alpha}(T, \boldsymbol{\alpha}). \quad (170)$$

The corresponding total reaction heat is

$$\dot{Q}_{\text{gen}} = \dot{Q}_{\text{gen}}(T, \boldsymbol{\alpha}). \quad (171)$$

The thermal model evaluates

$$\dot{T} = \frac{\dot{Q}_{\text{gen}} - hA(T - T_{\infty})}{C_{\text{th}}}. \quad (172)$$

9.4 Gas-generation contribution

If a gas-generation model is present, the reaction rates are also used to determine species-generation rates,

$$\dot{\mathbf{n}}_{\text{gen}} = \mathbf{f}_{\text{gas}}(\dot{\boldsymbol{\alpha}}). \quad (173)$$

Before vent opening,

$$\dot{\mathbf{n}} = \dot{\mathbf{n}}_{\text{gen}}. \quad (174)$$

9.5 Pressure reconstruction

Internal pressure is not integrated as an independent state variable.

Instead, it is reconstructed algebraically from the instantaneous gas inventory and temperature,

$$P = P(T, \mathbf{n}). \quad (175)$$

For the current ideal-gas formulation,

$$P = \frac{RT \sum_s n_s}{V_{\text{free}}}. \quad (176)$$

Avoiding pressure as an additional integrated variable ensures direct consistency between pressure and gas inventory.

9.6 Vent-opening event

When venting is enabled, the closed-cell integration defines an event function

$$g_{\text{vent}}(t, \mathbf{y}) = P(T, \mathbf{n}) - P_{\text{vent}}. \quad (177)$$

Vent opening occurs when

$$g_{\text{vent}} = 0. \quad (178)$$

The event terminates the closed-cell integration.

9.7 Two-phase integration

The current solver represents irreversible vent opening using two successive numerical integrations.

The first phase solves the closed-cell system,

$$\dot{\mathbf{y}} = \mathbf{F}_{\text{closed}}(t, \mathbf{y}), \quad (179)$$

until either the final time or a terminating event is reached.

If the vent-opening event occurs, the state at the event time

$$\mathbf{y}_{\text{vent}} \quad (180)$$

becomes the initial condition of the second phase.

The second integration solves

$$\dot{\mathbf{y}} = \mathbf{F}_{\text{open}}(t, \mathbf{y}), \quad (181)$$

where the gas balance is modified to include vent discharge,

$$\dot{\mathbf{n}} = \dot{\mathbf{n}}_{\text{gen}} - \dot{\mathbf{n}}_{\text{vent}}. \quad (182)$$

9.8 Solver flow

The numerical sequence is summarized in [figure 6](#).

9.9 Maximum-temperature event

A maximum permitted temperature may be specified as

$$T_{\text{max}}. \quad (183)$$

The corresponding event function is

$$g_T = T - T_{\text{max}}. \quad (184)$$

The integration terminates when the event crosses zero in the prescribed direction.

9.10 Maximum-pressure event

Similarly, a maximum-pressure event is represented by

$$g_P = P - P_{\text{max}}. \quad (185)$$

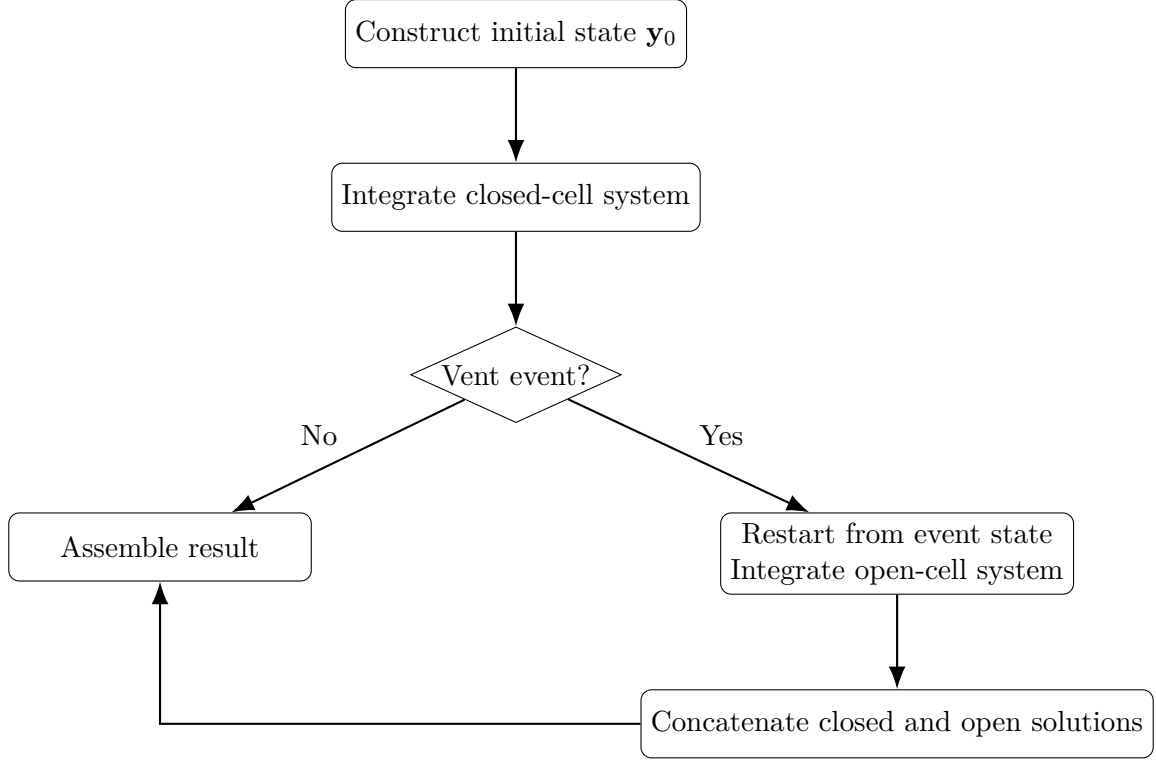


Figure 6: Numerical integration strategy used for irreversible vent opening in the current LiionTR solver.

This event can terminate either the closed or open phase of the simulation.

9.11 Stiff numerical integration

Thermal runaway kinetics frequently contain strongly temperature-sensitive Arrhenius terms spanning several orders of magnitude.

The coupled equations can consequently become stiff, particularly during the rapid acceleration preceding thermal runaway.

The default LiionTR solver therefore uses the backward differentiation formula method available through SciPy's `solve_ivp` interface [4].

The BDF method is an implicit variable-order integration method suitable for stiff ordinary differential equations.

9.12 Adaptive time stepping

The solver automatically adjusts its internal time step according to local error estimates.

The user may specify relative and absolute tolerances,

$$\text{rtol} \tag{186}$$

and

$$\text{atol}, \tag{187}$$

as well as an optional maximum time step.

The internal time-step size can become substantially smaller during the rapid thermal runaway phase than during the initial slow-heating period.

9.13 State reconstruction

After numerical integration, the resulting solver state is separated into physically meaningful output quantities.

These include

- time;
- cell temperature;
- individual reaction conversions;
- reaction progress rates;
- heat-generation rates;
- gas-species inventories;
- total gas amount;
- gas composition;
- pressure;
- vent status.

For vented simulations, the two numerical phases are concatenated into a single continuous simulation result.

9.14 Numerical architecture

The numerical solver acts as the orchestration layer between the individual physical models.

Conceptually,

$$\mathbf{y} \longrightarrow \left\{ \begin{array}{l} \text{reaction model,} \\ \text{thermal model,} \\ \text{gas model,} \\ \text{pressure model,} \\ \text{vent model} \end{array} \right. \longrightarrow \dot{\mathbf{y}}. \quad (188)$$

Individual physical components therefore remain independent of the specific numerical integration algorithm.

9.15 Current limitations

The current numerical formulation assumes:

- a single lumped temperature state;
- scalar reaction conversions;
- species-resolved gas inventories;
- pressure reconstructed algebraically;
- instantaneous transition from closed to fully open vent;
- no resealing after vent opening;
- no differential-algebraic constraints beyond algebraic reconstruction of derived quantities;
- no spatial discretization.

9.16 Future numerical extensions

Future developments may include:

- alternative stiff and non-stiff integrators;
- explicit Jacobian evaluation;
- sparse Jacobian support;
- differential-algebraic formulations;
- adaptive vent-opening models;
- integration of internal energy instead of temperature;
- spatial thermal discretization;
- cell-to-cell propagation systems;
- sensitivity equations and parameter estimation.

10 Verification

Verification of LiionTR focuses on determining whether the implemented numerical models reproduce their intended mathematical formulations.

This distinction is important:

- *verification* evaluates whether the equations are solved and implemented correctly;
- *validation* evaluates whether those equations reproduce physical experimental behavior.

The current development stage places primary emphasis on verification.

10.1 Unit-level verification

Individual LiionTR components are tested independently wherever possible.

The test suite includes checks for:

- geometric quantities;
- material properties;
- cell mass and thermal capacity;
- kinetic-rate calculations;
- reaction-progress laws;
- reaction heat generation;
- reaction-network aggregation;
- gas inventory properties;
- gas-generation rates;
- ideal-gas pressure;
- choked and unchoked vent flow;
- event handling;
- coupled numerical simulations.

Unit-level testing is particularly important because the modular architecture permits most physical components to be evaluated without running a complete thermal runaway simulation.

10.2 Dimensional consistency

All internal physical quantities are represented in SI units.

Typical units include

$$T : \text{K}, \tag{189}$$

$$P : \text{Pa}, \tag{190}$$

$$E_a : \text{J/mol}, \tag{191}$$

$$\Delta H : \text{J/kg}, \tag{192}$$

$$W : \text{kg/m}^3, \tag{193}$$

$$\dot{Q} : \text{W}, \tag{194}$$

$$n : \text{mol}. \tag{195}$$

Literature parameters reported in alternative units are explicitly converted before construction of the corresponding LiionTR models.

10.3 Reaction-energy consistency

For a reaction with effective mass fraction w , reaction enthalpy ΔH , and conversion change

$$\Delta\alpha = \alpha_f - \alpha_0, \quad (196)$$

the expected specific energy release relative to total cell mass is

$$q_{\text{expected}} = w\Delta H\Delta\alpha. \quad (197)$$

For a complete reaction network,

$$q_{\text{expected,total}} = \sum_i w_i \Delta H_i \Delta\alpha_i, \quad (198)$$

with the appropriate extension for multi-channel reactions.

Numerically integrating the heat-generation rate provides

$$q_{\text{integrated}} = \frac{1}{m_{\text{cell}}} \int \dot{Q}_{\text{gen}} dt. \quad (199)$$

Agreement between [equations \(198\)](#) and [\(199\)](#) provides an important consistency check between reaction progress and heat generation.

10.4 Hu model energy check

A representative numerical simulation of the implemented Hu et al. reaction network was performed from an elevated initial temperature in order to activate the thermal runaway mechanism.

For the tested case, the integrated specific reaction energy was

$$q_{\text{integrated}} \approx 386.6 \text{ kJ/kg}. \quad (200)$$

The integrated heat release was consistent with the energy obtained from the change in reaction states and the corresponding reaction enthalpies.

This test verifies the internal consistency of the implemented reaction network and heat-generation calculation. It should not be interpreted as experimental validation of the Hu model.

10.5 Thermal-response consistency

For an adiabatic system with constant heat capacity and no heat loss,

$$C_{\text{th}} \frac{dT}{dt} = \dot{Q}_{\text{gen}}. \quad (201)$$

Integrating gives

$$\Delta T = \frac{Q_{\text{gen}}}{C_{\text{th}}}. \quad (202)$$

This relationship provides a direct verification target for simplified thermal problems.

10.6 Gas-generation conservation

For a closed system with no venting, the numerically integrated gas inventory should agree with the cumulative reaction-yield relation.

For species s ,

$$n_s(t) - n_s(0) = \sum_i Y_{i,s} m_{\text{cell}} w_i [\alpha_i(t) - \alpha_i(0)]. \quad (203)$$

Agreement verifies consistency between reaction progress, gas-generation rates, and integrated species inventories.

10.7 Pressure verification

For a closed ideal-gas problem, pressure reconstructed from the numerical state must satisfy

$$P = \frac{n_{\text{tot}} RT}{V_{\text{free}}}. \quad (204)$$

Synthetic problems with constant temperature or constant gas amount can therefore be compared directly against analytical pressure evolution.

10.8 Vent-flow verification

The venting implementation is tested separately in both compressible-flow regimes.

For

$$\frac{P_d}{P_u} \leq r_{\text{crit}}, \quad (205)$$

the choked-flow expression must be selected.

For

$$\frac{P_d}{P_u} > r_{\text{crit}}, \quad (206)$$

the unchoked-flow expression must be selected.

The model also verifies that

$$P_u \leq P_d \quad (207)$$

produces zero outward discharge.

10.9 Vented coupled simulation

A synthetic coupled simulation has been used to verify the complete sequence

$$\text{reaction} \rightarrow \text{heat} \rightarrow T \rightarrow \text{gas generation} \rightarrow P \rightarrow \text{vent opening} \rightarrow \text{gas discharge}. \quad (208)$$

The test demonstrates that:

1. reaction progress generates heat;
2. heat generation increases temperature;
3. reaction progress generates gas;
4. gas accumulation and heating increase pressure;
5. the vent opens when the prescribed pressure is reached;
6. the solver transitions irreversibly to the open-vent system;
7. gas discharge reduces the internal inventory;
8. pressure approaches the downstream pressure after the dominant gas-generation period.

This synthetic problem is designed to verify model coupling and event handling rather than reproduce a particular commercial battery cell.

10.10 Regression testing

Automated regression tests are used to detect unintended changes in physical behavior as the codebase evolves.

This is particularly important for coupled models because a local software modification can alter:

- reaction rates;
- integrated reaction energy;
- event times;
- maximum temperature;
- pressure evolution;
- vent discharge.

Regression tests therefore complement analytical unit tests.

10.11 Verification hierarchy

The current verification strategy follows the hierarchy illustrated in [figure 7](#).

10.12 Experimental validation

Experimental validation remains an essential future step.

Relevant validation data may include:

- differential scanning calorimetry measurements;
- accelerating rate calorimetry measurements;

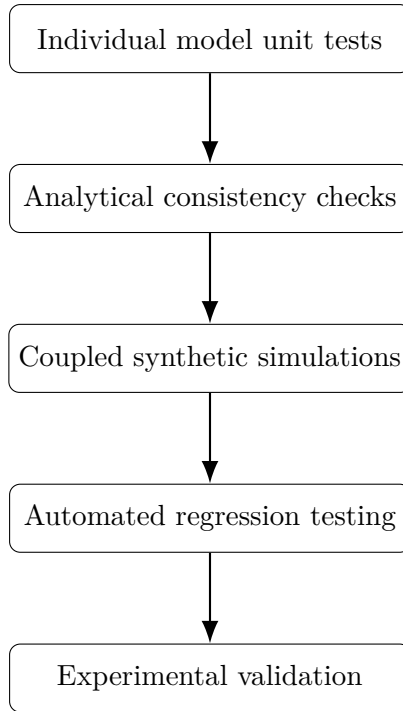


Figure 7: Verification and validation hierarchy intended for LiionTR. Experimental validation represents a future development stage following verification of the numerical implementation.

- cell-surface temperature histories;
- pressure measurements;
- vent-opening times;
- total vented gas quantities;
- gas-composition measurements;
- abuse-test thermal runaway onset temperatures.

The modular structure of LiionTR is intended to allow kinetic, thermochemical, pressure, and venting submodels to be calibrated and validated independently before complete-cell validation is attempted.

11 Reference Simulation Results

The behavior of the implemented thermal runaway framework is illustrated using a reference simulation based on the reaction network of Hu et al. [1].

The purpose of this simulation is to demonstrate the coupled response of the implemented reaction and thermal models. The calculation is not intended as an independent experimental validation of the original model.

11.1 Reference simulation conditions

The simulation is initialized at an elevated temperature in order to activate the thermal runaway reaction network within a short numerical time interval.

The principal simulation conditions are summarized in [table 3](#).

Table 3: Conditions used for the Hu et al. reference simulation.

Parameter	Value	Unit
Initial temperature	480	K
Simulation duration	20	s
Convective heat-transfer coefficient	10	W/(m ² K)
Maximum permitted temperature	1200	K

11.2 Temperature evolution

The simulated temperature evolution is shown in [figure 8](#).

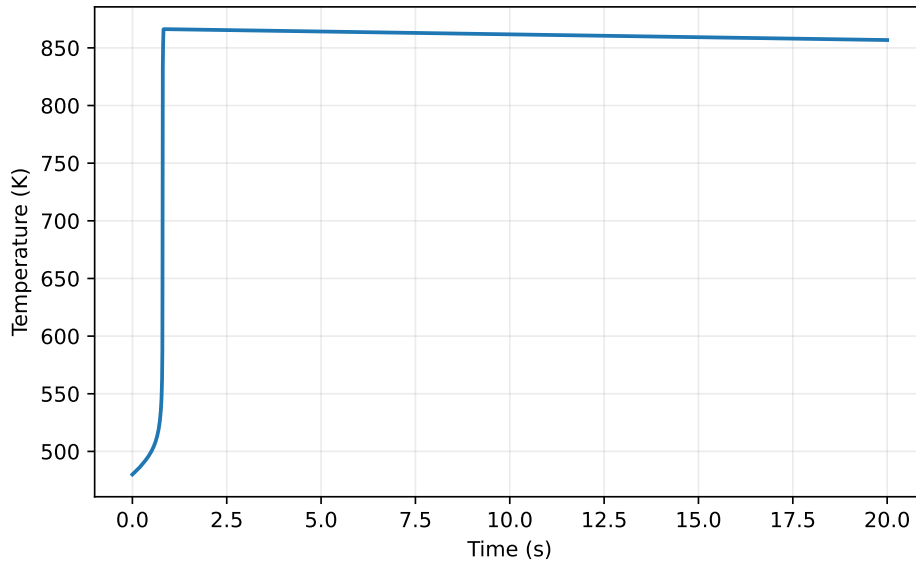


Figure 8: Cell temperature predicted by LiionTR for the Hu et al. reference thermal runaway simulation.

11.3 Reaction progression

The evolution of the four reaction conversions is shown in [figure 9](#).

11.4 Reaction heat generation

The individual reaction contributions to heat generation are shown in [figure 10](#).

11.5 Integrated reaction energy

The total specific reaction energy is calculated from

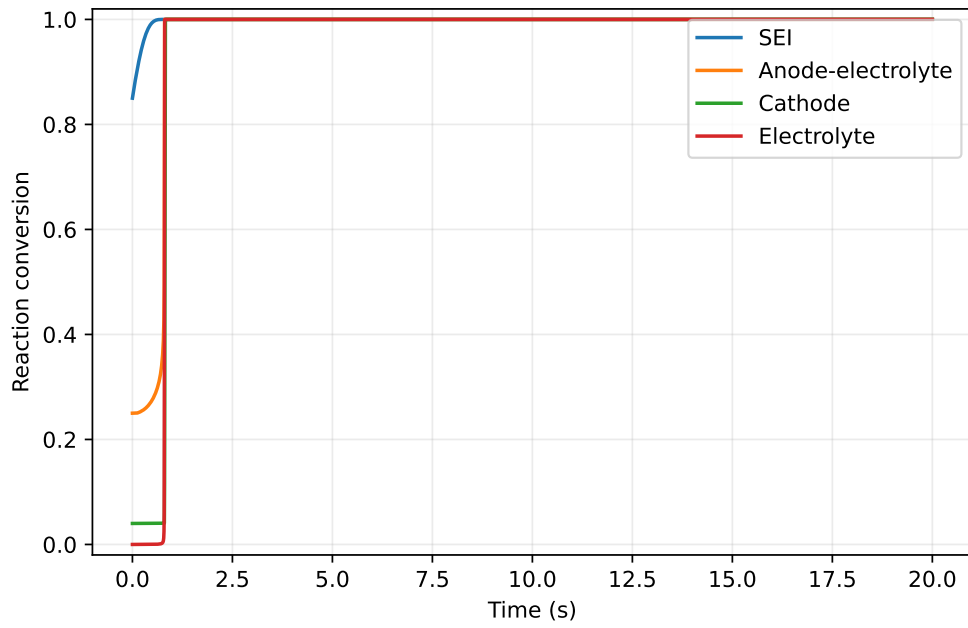


Figure 9: Evolution of the SEI, anode-electrolyte, cathode, and electrolyte reaction conversions during the reference simulation.

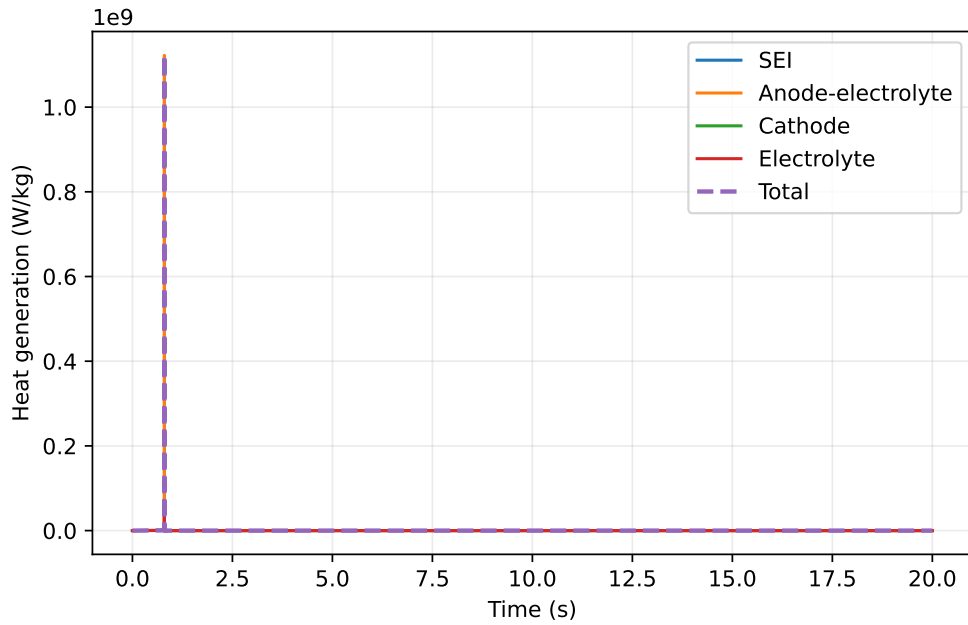


Figure 10: Mass-specific heat-generation rates associated with the thermal runaway reactions in the Hu et al. reference simulation.

$$q_{\text{reaction}} = \int \dot{q}_{\text{total}} dt. \quad (209)$$

For the present reference calculation, the integrated reaction energy is approximately

$$q_{\text{reaction}} = 386.6 \text{ kJ/kg}. \quad (210)$$

The final simulated cell temperature is approximately

$$T_f = 856.8 \text{ K}. \quad (211)$$

These values characterize the selected numerical reference case and should not be interpreted as general thermal runaway characteristics of lithium-ion cells.

12 Future Development

LiionTR is currently a reduced-order thermal runaway framework with explicit models for reaction kinetics, heat generation, gas generation, pressure, and venting.

Several extensions are required before the framework can represent the full range of physical processes relevant to lithium-ion battery failure.

Development priorities are described below.

12.1 Thermochemical gas modeling

The current gas-generation formulation uses prescribed species yields.

This approach is useful for empirical models and verification but cannot predict gas composition from the chemical state of the reacting battery materials.

A future thermochemical formulation will represent reaction products in terms of elemental inventories.

Conceptually,

$$\text{thermal decomposition} \rightarrow \text{element source terms} \rightarrow \text{element inventory} \rightarrow \text{thermochemical equilibrium}. \quad (212)$$

Relevant elements may include

$$\text{C, H, O, N, F, P, Li}. \quad (213)$$

A thermochemical equilibrium backend can then determine gas composition subject to specified thermodynamic constraints.

12.2 Cantera integration

Cantera is a strong candidate for the thermochemical backend.

A future LiionTR interface may provide a generic abstraction such as

$$\text{GasEquilibriumBackend}, \tag{214}$$

with a Cantera implementation responsible for

- equilibrium gas composition;
- mixture molar mass;
- heat capacities;
- enthalpy and internal energy;
- heat-capacity ratio;
- other thermodynamic properties required by venting.

The initial implementation may use constant-volume, constant-temperature equilibrium calculations.

More advanced formulations may require coupled energy-conserving equilibrium calculations.

12.3 Energy-consistent thermochemistry

The current thermal state is temperature.

This is convenient for reduced-order calculations but may become insufficient when significant energy is redistributed through chemical equilibrium, vaporization, gas discharge, or phase change.

A future formulation may therefore evolve internal energy,

$$U, \tag{215}$$

rather than temperature directly.

Temperature would then be reconstructed from the thermodynamic state,

$$T = T(U, V, \mathbf{n}). \tag{216}$$

Such a formulation would provide a more rigorous basis for energetic coupling between condensed reactions and gas-phase thermochemistry.

12.4 Electrochemical coupling

Thermal runaway behavior depends strongly on the electrochemical state of the cell.

Relevant quantities include

- state of charge;
- electrode lithium concentration;
- electrode potential;

- current;
- irreversible electrochemical heating;
- reversible entropic heating.

PyBaMM may eventually provide these quantities to LiionTR.

The intended coupling is not to replace PyBaMM electrochemistry, but to allow electrochemical state variables to modify thermal runaway kinetics, reaction inventories, or onset conditions.

12.5 Calorimetric parameter estimation

Thermal runaway kinetic parameters are often derived from differential scanning calorimetry or accelerating rate calorimetry measurements.

LiionTR may therefore include tools for fitting model parameters to experimental data.

Potential parameters include

$$A, \quad E_a, \quad \Delta H, \quad n, \quad \alpha_0. \quad (217)$$

Parameter estimation could be formulated as an optimization problem,

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}), \quad (218)$$

where $\boldsymbol{\theta}$ is the parameter vector and J measures the difference between simulation and experiment.

12.6 Sensitivity and uncertainty analysis

Published thermal runaway parameters can vary substantially between cells, chemistries, measurement methods, and states of charge.

Future development should therefore consider parameter uncertainty.

Relevant outputs may include sensitivities of

- thermal runaway onset time;
- maximum temperature;
- maximum pressure;
- vent-opening time;
- total gas generation;
- total energy release.

Both local sensitivity analysis and Monte Carlo uncertainty propagation may be useful.

12.7 Temperature-dependent material properties

The current cell thermal properties are effectively constant.

Future material models may support

$$c_p = c_p(T), \quad (219)$$

$$k = k(T), \quad (220)$$

and potentially

$$\rho = \rho(T). \quad (221)$$

Phase transitions and latent-heat effects may also be relevant at severe thermal runaway temperatures.

12.8 Spatial thermal models

The present lumped formulation cannot reproduce internal temperature gradients.

A natural extension is a one-dimensional radial model for cylindrical cells,

$$\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \dot{q}_{\text{gen}}. \quad (222)$$

More general finite-volume or finite-element formulations could later support cylindrical, prismatic, and pouch cells.

12.9 Cell-to-cell propagation

Battery-module safety analysis requires coupling between multiple cells.

For cell i , a generic thermal balance may be written as

$$C_i \frac{dT_i}{dt} = \dot{Q}_{i,\text{gen}} - \dot{Q}_{i,\text{ambient}} + \sum_j \dot{Q}_{j \rightarrow i}. \quad (223)$$

The inter-cell term

$$\dot{Q}_{j \rightarrow i} \quad (224)$$

may include conduction, radiation, convection, or heat transfer from vent gases and flames.

The modular cell representation already provides a basis for this extension.

12.10 Mechanical venting and rupture

The current vent-opening model uses a prescribed pressure threshold and instantaneous transition to a fixed vent area.

More realistic models may include

- progressive vent displacement;

- pressure-dependent vent area;
- burst-disk failure;
- casing deformation;
- rupture;
- multiple vent paths.

A mechanical model could therefore determine

$$A_v = A_v(P, T, t) \quad (225)$$

instead of treating vent area as constant.

12.11 Multiphase venting

Real battery vent products may contain gases, vapor, droplets, aerosols, and solid particles.

The current single-phase ideal-gas discharge model does not represent these phenomena. Future formulations may require

- electrolyte vaporization;
- flashing flow;
- liquid entrainment;
- aerosol transport;
- multiphase choking criteria.

12.12 External vent-jet and combustion models

Once material leaves the cell, additional physical processes may occur outside the battery.

These include

- turbulent jet formation;
- mixing with ambient air;
- ignition;
- flame propagation;
- radiant heat transfer;
- toxic-species transport.

These processes are outside the current LiionTR domain but could eventually be represented through interfaces to combustion or computational-fluid-dynamics tools.

12.13 Validation database

A robust thermal runaway framework requires systematic comparison with experimental measurements.

A future validation database could contain

- battery-cell metadata;
- chemistry;
- geometry;
- nominal capacity;
- state of charge;
- abuse condition;
- ARC data;
- DSC data;
- temperature histories;
- pressure histories;
- vent gas composition;
- vented mass;
- photographs or high-speed imaging metadata.

Standardized datasets would permit repeatable comparison between alternative LiionTR models.

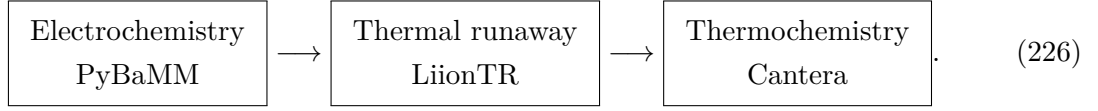
12.14 Software development

In addition to physical-model development, several software improvements are anticipated:

- stable public APIs;
- structured serialization of simulations;
- model configuration files;
- reproducible simulation metadata;
- plotting and post-processing utilities;
- parameter databases;
- expanded example simulations;
- automated benchmarking;
- optional parallel parameter studies.

12.15 Long-term architecture

The long-term architecture can be summarized conceptually as



LiionTR is intended to remain the layer responsible for coordinating battery-safety-specific physics, thermal runaway reactions, gas generation, pressure, venting, and propagation.

The framework should therefore remain modular enough that increasing the fidelity of one physical subsystem does not require redesigning the complete simulation architecture.

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